

7/11/14

07/11/14(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
Semester: July / January 2014 - November / April 2015

Max. Marks: 100

Duration: 3 Hrs

End Semester Exam

Class: SE

Semester: III

Name of the Course: Digital Logic Design and Analysis

Branch: COMP

Course Code: UCEC304

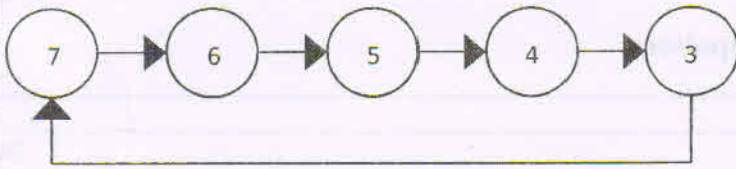
Instructions:

(1) All Questions are Compulsory

Question No.		Max. Marks	CO Mapped
Q1	Define the following terms: 1. Noise Immunity 2. Noise Margin 3. Fan-in & Fan-out 4. Propagation Delay 5. TTL CMOS logic levels	10	3
Q2(a)	1. Convert $(4CE)_{16}$ to its equivalent Octal form. 2. Convert $(0.122)_{10}$ to its equivalent Hex form. 3. Convert $(19)_{10}$ to BCD, XS-3 and Gray	10	1
Q2(b)	Represent $(89)_{10}$ in: 1. 1's complement representation 2. 2's complement representation	5	1
Q2(c)	Write short note on Gray Code and its applications (OR) Write short note on Error Correcting Code	5	1

$$\rightarrow [(A + \bar{A}B)(A + \bar{A}\bar{B})] [(CD + \bar{C}\bar{D}) + (C\bar{D} + \bar{C}D)] = A$$

Q3(a)	Prove the following using Boolean theorems: $[(A + AB)(A + \bar{A}B)] [(CD + \bar{C}\bar{D}) + (C\bar{D} + \bar{C}D)] = A$	5	2
Q3(b)	Simplify the following expression and realize using 2 input gates : $f(A,B,C,D) = \sum m(4, 5, 8, 9, 11, 12, 13, 15)$ (OR) $F(A,B,C,D) = \pi M(4, 6, 8, 9, 10, 12, 13, 14) + d(0, 2, 5)$	5	2
Q3(c)	Minimize the following expression using Quine McCluskey tabular method. $f(A,B,C,D) = \sum m(2, 9, 10, 11, 13, 15)$	10	2
Q4(a)	Convert T Flip-flop to D flip-flop (OR) Implement 8-bit adder using 4-bit full adder.	6	2
Q4(b)	Explain what is race around condition in J-K flip-flop. How this problem could be solved?	4	2

	(OR) Implement using Suitable Demux : $f_1(AB,C) = \sum (0,1,3,7) + d(2,5)$ $f_2(AB,C) = \sum (1,5,7)$ $f_3(AB,C) = \sum (0,1,3,7)$		
Q4(c)	Design ripple counter for state diagram shown in the fig.  (OR) Draw a logic diagram of 4 bit bi-directional shift register using D flip-flop using D flip-flops. Explain its working.	10	2
Q4(c)	Using J-K flip-flops realize a three bit Gray code counter. Use K-maps for reductions	10	2
Q5(a)	Distinguish between Entity declaration & architecture declaration	5	4
Q5(b)	Write short note on Behavioral modelling (OR) Write short note on Data flow modelling	5	4
Q5(c)	Compare FPGA with CPLD (OR) Explain the architecture and working of FPGA	10	4